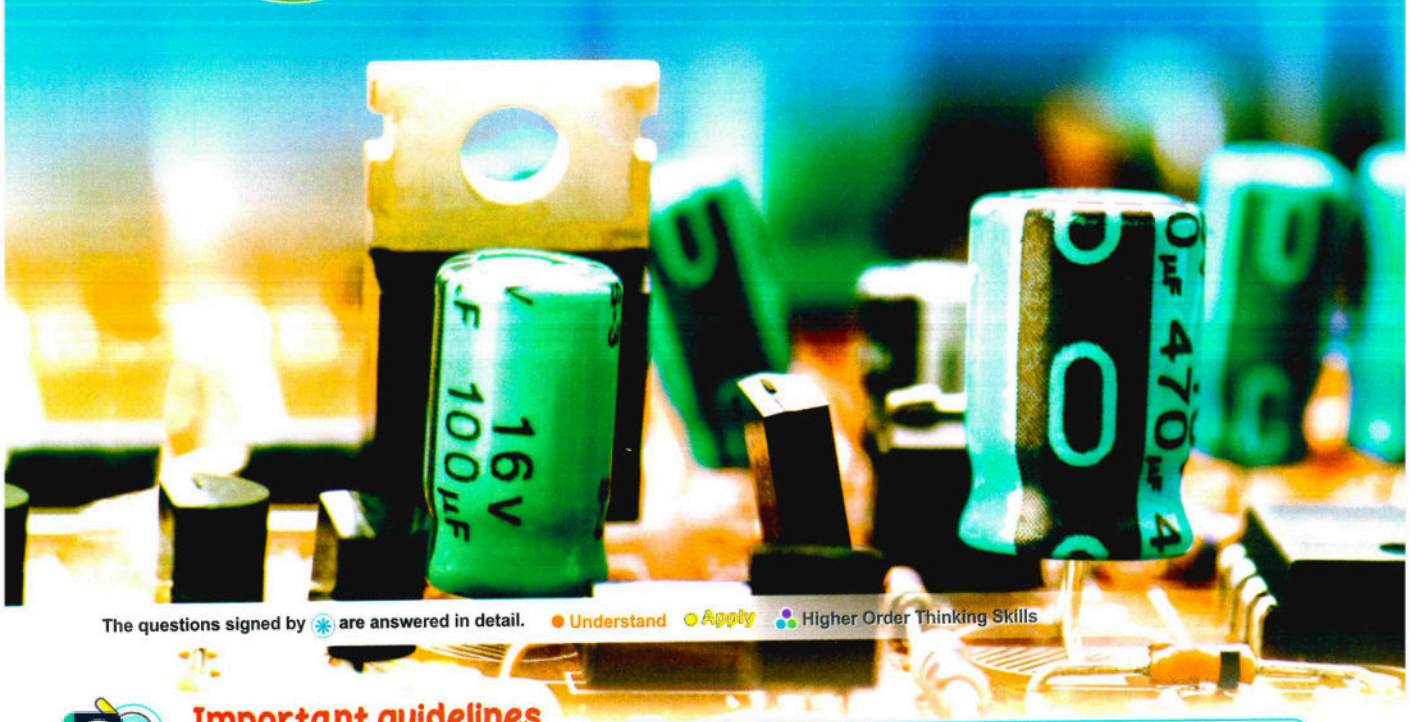
Chapter 8

Questions on Lesson Two

• The Transistor.
• Digital and Analog Electronics.

Interactive test



The questions signed by ✻ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

⇒ **Transistor as an amplifier:**

- To determine the emitter current (I_E): $I_E = I_C + I_B$
- To determine the current gain (β_e): $\beta_e = \dfrac{I_C}{I_B} = \dfrac{\alpha_e}{1 - \alpha_e}$
- To determine the current transfer ratio (α_e): $\alpha_e = \dfrac{I_C}{I_E} = \dfrac{\beta_e}{1 + \beta_e}$

⇒ **Transistor as a switch:**

- To determine the battery voltage (V_{CC}): $V_{CC} = V_{CE} + I_C R_C$

First Multiple choice questions

The transistor

1 The base region in the transistor is relatively .....

(a) thick and has higher concentration of impurities
(b) thin and has higher concentration of impurities
(c) thick and has lower concentration of impurities
(d) thin and has lower concentration of impurities

2 In a pnp transistor, the dominant charge carriers in both of the emitter and the collector are ..... .

(a) acceptor ions (b) donor ions (c) free electrons (d) holes

3 The npn transistor consists of ..... .

(a) one pn junction (b) two pn junctions (c) three pn junctions (d) four pn junctions

4 The two opposite figures (1) and (2) represent transistor connection circuits in a common-emitter configuration. In which of them is the transistor used as a switch in the off state ?

(a) Figure (1). (b) Figure (2).

(c) Both figures. (d) Neither of them.

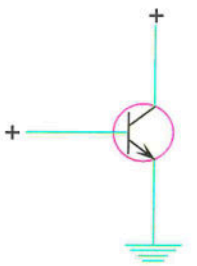


Figure (1)

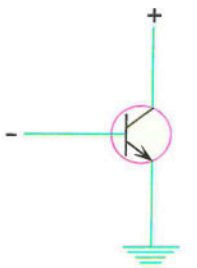


Figure (2)

5 In the transistor, the ratio of the concentration of impurities in the emitter to that in the collector is ..... .

(a) equal to one (b) greater than one (c) less than one (d) indeterminable

6 When connecting a transistor in the common base mode, if the current transfer ratio is α_e and current gain is β_e then ..... .

(a) $\alpha_e < 1$ (b) $\beta_e > 1$ (c) $\alpha_e = \frac{\beta_e}{1+\beta_e}$ (d) all the previous

7 Transistor npn is connected in a circuit in a common emitter configuration. If the base is given a high positive potential, the transistor acts as ..... .

(a) a half wave rectifier for current (b) a closed switch

(c) an opened switch (d) a full wave rectifier

8 If the current through the collector of an npn transistor is 18 mA and that represents 90% of the emitter current, so the emitter current (I_E) and the base current (I_B), respectively, are ..... .

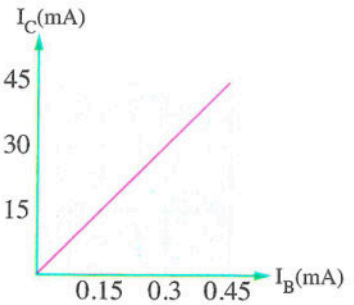
(a) 20 mA, 2 mA (b) 30 mA, 12 mA (c) 19 mA, 1 mA (d) 20 mA, 4 mA

9 The opposite graph represents the relation between the collector current (I_C) and the base current (I_B) in a pnp transistor, so;

(i) The value of α_e equals ..... .

(a) 0.96 (b) 0.98

(c) 0.99 (d) 1

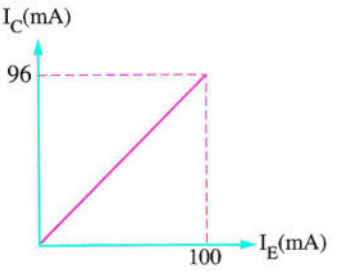


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(ii) The intensity of the emitter current, when $I_C = 45$ mA, equals .....

(a) 20.12 mA (b) 30.15 mA (c) 40.25 mA (d) 45.45 mA

10 * The opposite graph represents the relation between the collector current (I_C) and the emitter current (I_E) for an npn transistor, so the values of α_e and β_e, respectively, are .....



(a) 0.96, 24 (b) 0.96, 48

(c) 0.49, 32 (d) 0.49, 64

11 * If α_e for a transistor is 0.99 and the base current is 100 μA, then the current gain (β_e) and the collector current (I_C), respectively, equal ..... .

(a) 98, 9.8×10^{-3} A (b) 99, 9.9×10^{-3} A

(c) 99, 9.9 A (d) 98, 9.8 A

12 * If the base current in a transistor is 24 μA and its current gain is 24, then the current transfer ratio (α_e) and collector current (I_C), respectively, are ..... .

(a) 0.76, 576×10^{-3} A (b) 0.76, 576×10^{-6} A

(c) 0.96, 576×10^{-3} A (d) 0.96, 576×10^{-6} A

13 * When using a transistor as an amplifier, if the electrical signal into the base of the transistor is 200 μA and the collector current becomes 10 mA, then β_e and α_e, respectively, are ..... .

(a) 20, 0.95 (b) 20, 0.98 (c) 50, 0.95 (d) 50, 0.98

14 If the current gain in a transistor of npn type is 99 and the collector current is 10 mA, hence the current transfer ratio (α_e) and emitter current (I_E) respectively are ..... .

(a) 0.98, 10.1×10^{-3} A (b) 0.98, 0.1×10^{-3} A

(c) 0.99, 0.1×10^{-3} A (d) 0.99, 10.1×10^{-3} A

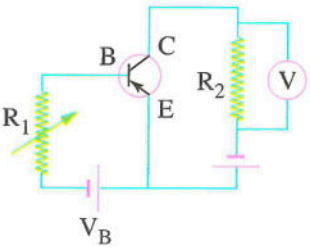
15 * In a transistor circuit in which ($V_{CC} = 1.5$ V), the potential difference between the collector and the emitter ($V_{CE} = 0.5$ V) and the resistance of the collector circuit ($R_C = 500\ \Omega$), so the collector current (I_C) equals ..... .

(a) 1×10^{-3} A (b) 2×10^{-3} A

(c) 3×10^{-3} A (d) 4×10^{-3} A

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16 In the opposite transistor circuit, when the resistance of R_1 increases, the reading of the voltmeter ..... .



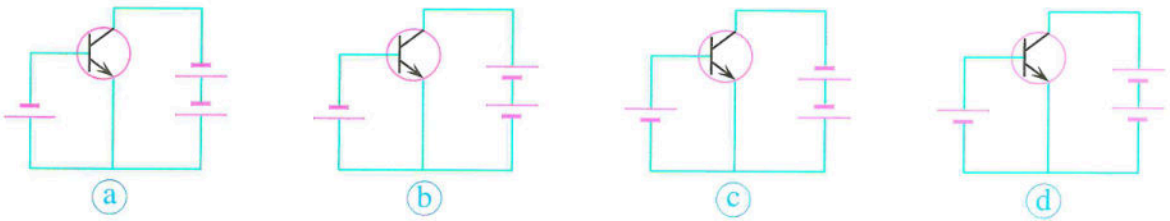
(a) increases

(b) decreases but doesn't reach zero

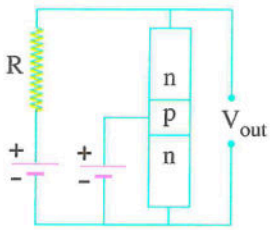
(c) remains constant

(d) equals zero

17 The circuit that shows the correct way of connecting the batteries in a transistor circuit in a common emitter mode of a closed switch is ..... .



18 In the shown transistor circuit, the transistor is connected as ..... .



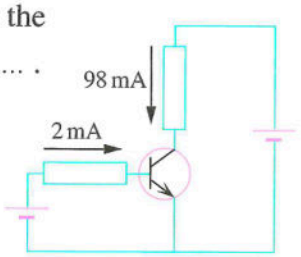
(a) an opened switch

(b) a common emitter

(c) a common base

(d) answer is indeterminable

19 * The opposite figure represents an npn transistor circuit, so from the given data, the two constants β_e and α_e, respectively, equal ..... .



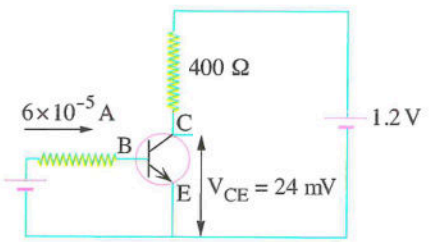
(a) 50, 0.98

(b) 49, 0.99

(c) 49, 0.98

(d) 50, 0.99

20 * The opposite figure represents an npn transistor circuit, so the current transfer ratio (α_e) equals ..... .



(a) 0.92

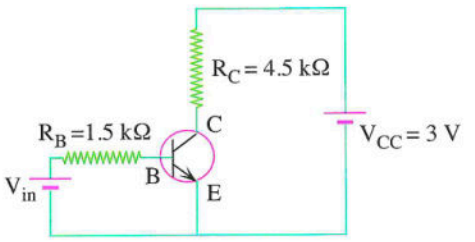
(b) 0.95

(c) 0.96

(d) 0.98

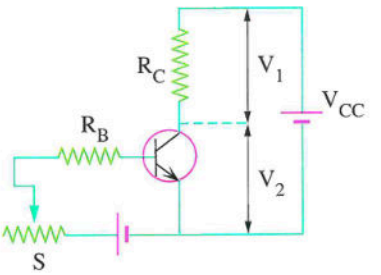
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21 The opposite diagram represents a circuit in which the transistor is used as a switch, so if the potential difference across the base resistor R_B equals 0.01 V and $\beta_e = 75$, then ..... .



	The intensity of the base current (I_B)	The output voltage (V_{CE})
(a)	3.33×10^{-6} A	0.55 V
(b)	3.33×10^{-6} A	0.75 V
(c)	6.67×10^{-6} A	0.55 V
(d)	6.67×10^{-6} A	0.75 V

22 The opposite figure represents a transistor (npn) circuit, where the transistor is switched on. When the resistance of the rheostat (S) increases, then the magnitudes of potential differences V_1 and V_2, respectively, will ..... .



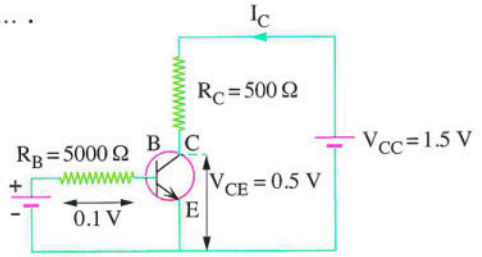
(a) decrease, decrease

(b) decrease, increase

(c) increase, increase

(d) increase, decrease

23 ✳ In the opposite circuit, the value of I_E equals ..... .



(a) 2×10^{-3} A

(b) 2.02×10^{-3} A

(c) 2×10^{-5} A

(d) 2.02×10^{-5} A

24 In npn transistor, if the difference between the intensities of the collector and base currents ($I_C - I_B$) equals 97 mA, and the intensity of the emitter current (I_E) equals 99 mA, so the current gain (β_e) equals ..... .

(a) 97 (b) 98 (c) 99 (d) 100

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25 In the shown transistor circuit in figure (1), the relation between the ammeter reading (I_C) and voltameter reading (V_{CE}) is represented graphically as shown in figure (2),

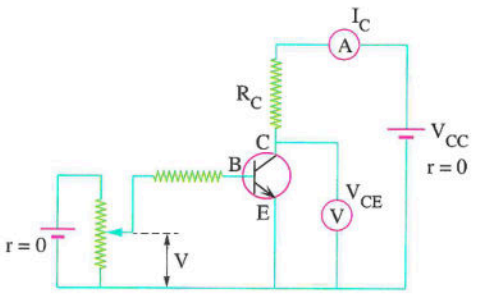


Figure (1)

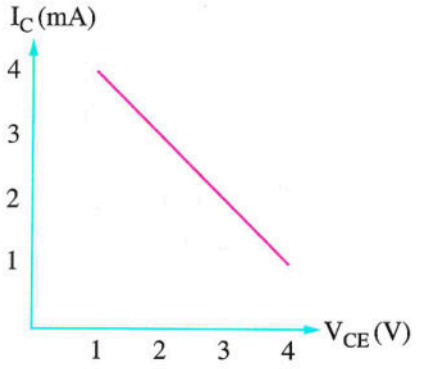


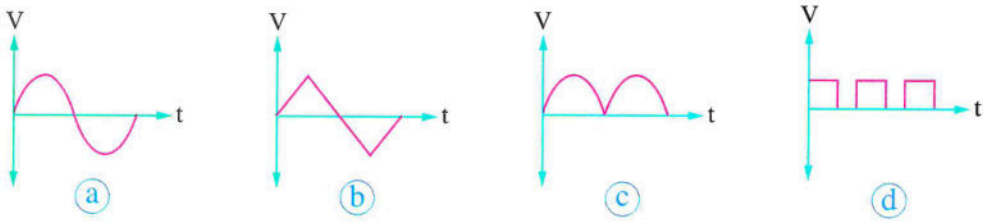
Figure (2)

then the values of R_C and V_{CC}, respectively, are ..... .

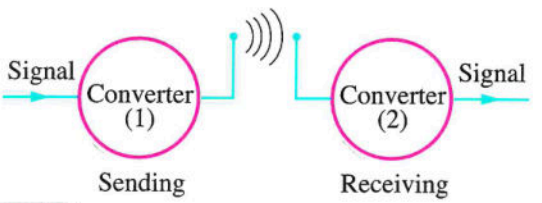
(a) 1 kΩ, 4 V (b) 1 kΩ, 5 V (c) 2 kΩ, 4 V (d) 2 kΩ, 5 V

Digital and analog electronic devices

26 Which of the following graphs represents the change of voltage (V) for a signal in a digital electronic device with time (t)?



27 In digital electronics, to send and receive signals that carry information, converter (1) is used at sending and converter (2) is used at receiving, then ..... .



	Converter (1)	Converter (2)
(a)	analog to digital	analog to digital
(b)	analog to digital	digital to analog
(c)	digital to analog	analog to digital
(d)	digital to analog	digital to analog

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28 Which of the following statements correctly represents the order of the stages of digital transmission and reception?

(a) Physical quantity ⟶ Analog signal ⟶ Digital signal ⟶ Analog signal ⟶ Physical quantity

(b) Physical quantity ⟶ Digital signal ⟶ Analog signal ⟶ Digital signal ⟶ Physical quantity

(c) Analog signal ⟶ Physical quantity ⟶ Digital signal ⟶ Physical quantity ⟶ Analog signal

(d) Digital signal ⟶ Analog signal ⟶ Physical quantity ⟶ Analog signal ⟶ Digital signal

29 The code for the decimal number 20 according to the binary system is ..... .

(a) $(10101)_2$ (b) $(10100)_2$ (c) $(11100)_2$ (d) $(00111)_2$

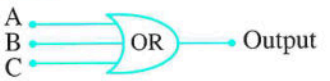
30 The binary number which is equivalent to the summation of the following decimal numbers (1 + 1 + 1 + 1) is ..... .

(a) $(1111)_2$ (b) $(101)_2$ (c) $(110)_2$ (d) $(100)_2$

31 The decimal number which is equivalent to the binary number $(1010)_2$ is ..... .

(a) 4 (b) 8 (c) 10 (d) 12

32 The probability of getting an output of (1) in the opposite logic gate is ..... .



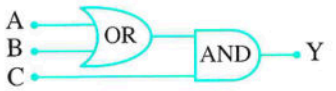
(a) 10 % (b) 20 % (c) 50.9 % (d) 87.5 %

33 The opposite figure shows a logic gate, so the number of probable combinations that gives high value at the output equals ..... .



(a) 0 (b) 1 (c) 2 (d) 3

34 In the shown logic gates, to get an output Y = 1, the inputs (A, B, C) have to be .....



	A	B	C
(a)	0	1	0
(b)	0	0	1
(c)	1	0	1
(d)	1	0	0

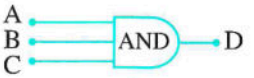
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35 Which of the following logic gates matches the opposite truth table?

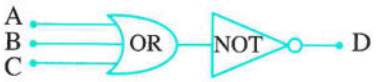
A	B	C	D
0	0	1	1
0	1	0	1
0	1	1	1



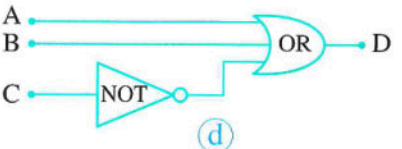
(a)



(b)



(c)

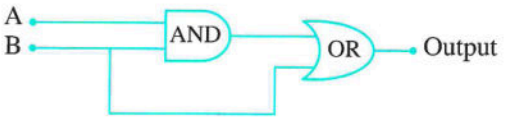


(d)

36 When digital signal $(11001)_2$ is applied at the input of a NOT gate, the output signal will be ..... .

(a) $(00111)_2$ (b) $(00110)_2$ (c) $(10101)_2$ (d) $(11001)_2$

37 In the opposite logic circuit, if the inputs are as shown in the opposite table, so the outputs will be ..... .



A	B
0	0
1	1

Output
0
1

(a)

Output
1
1

(b)

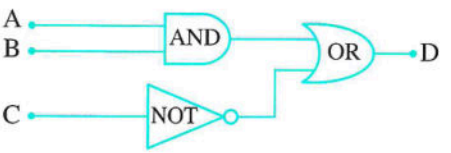
Output
0
0

(c)

Output
1
0

(d)

38 In the shown logic circuit, which of the following inputs gives the output D = 1?



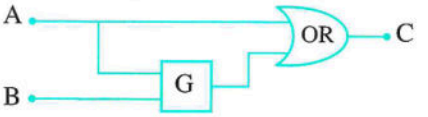
(a) A = 0, B = 0, C = 1

(b) A = 1, B = 0, C = 1

(c) A = 1, B = 0, C = 0

(d) A = 0, B = 1, C = 1

39 The opposite figure shows a logic circuit with its truth table, so the logic gate G is ..... .



(a) AND

(b) OR

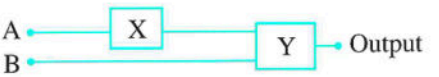
(c) NOT

(d) AND or OR

A	B	C
0	0	0
0	1	0
1	0	1
1	1	1

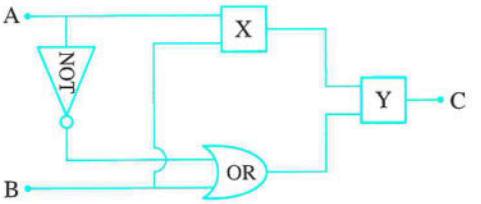
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40 The opposite figure shows a logic circuit, if the output of this circuit becomes (1) when its inputs on A and B are (0, 0), the components X and Y, respectively, are ..... .



(a) OR, AND (b) NOT, AND (c) NOT, NOT (d) NOT, OR

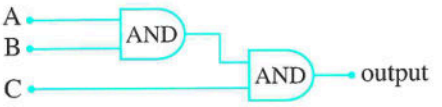
41 The opposite figure shows a logic circuit and its truth table, so the two logic gates X and Y are ..... .



	Gate X	Gate Y
(a)	OR	AND
(b)	AND	AND
(c)	OR	OR
(d)	AND	OR

A	B	C
0	0	1
0	1	1
1	0	0
1	1	1

42 The opposite figure shows a logic circuit, and the truth table that follows the circuit shows some input and output values for this circuit, so the values of x, y, and z in the truth table are ..... .



input			output
A	B	C	
x	1	1	0
1	1	0	y
1	z	1	1

(a)

x	y	z
0	0	0

(b)

x	y	z
1	0	0

(c)

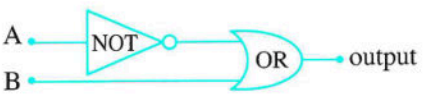
x	y	z
0	0	1

(d)

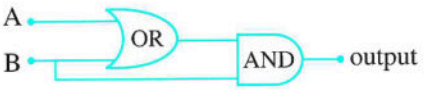
x	y	z
0	1	1

43 Which of the following logic circuits matches the opposite truth table?

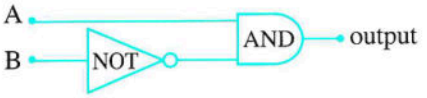
A	B	output
0	0	1
0	1	1
1	0	0
1	1	1



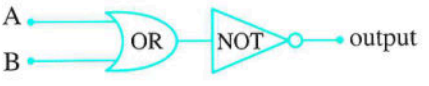
(a)



(b)



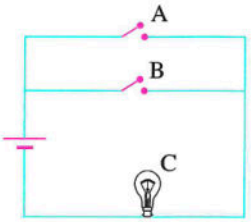
(c)



(d)

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44 In the opposite electric circuit, if the two switches (A) and (B) represent the inputs and the lightbulb (C) represents the output, the correct truth table for this circuit is ..... .



A	B	C
0	1	1
1	1	0

(a)

A	B	C
0	1	0
1	1	1

(b)

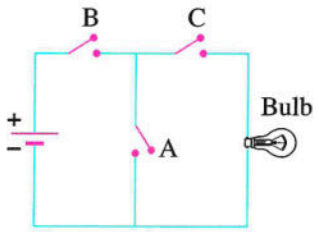
A	B	C
1	0	1
0	1	1

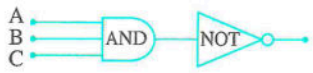
(c)

A	B	C
1	0	0
0	1	0

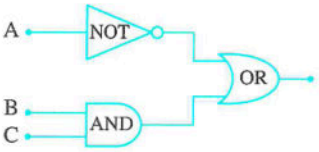
(d)

45 Which of the following logic gates represents the opposite electric circuit?

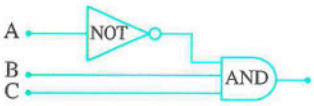




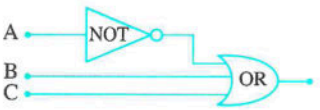
(a)



(b)



(c)



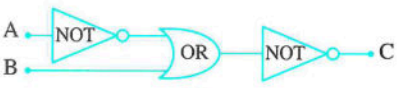
(d)

46 In which of the following logic gates will the output be equivalent to the decimal number 11, if the inputs are listed in the opposite table?

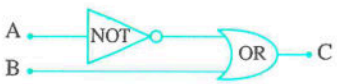
A	B
0	0
0	1
1	0
1	1



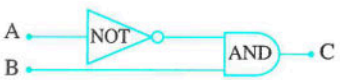
(a)



(b)



(c)

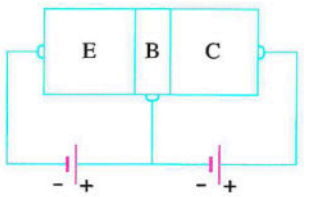


(d)

Second Essay questions

The transistor

1 **Explain what happens to** the free electrons in each of the regions of an npn transistor when it is connected as shown in the opposite figure.

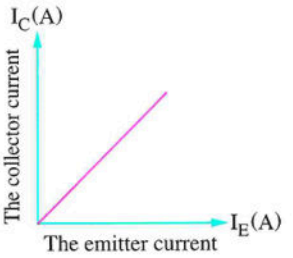


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2 **Give reasons for:** In the transistor, the current transfer ratio (α_e) is close to one while the current gain (β_e) is greater than one.

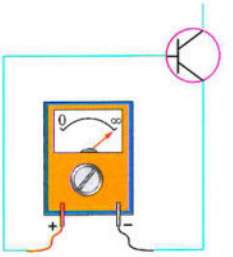
3 **Compare between** the emitter and the collector in the npn transistor in the case of a common base connection (in terms of : valency of impurities - type of connection with the base - the potential barrier with the base).

4 **Write the mathematical equation that relates the two variables in the opposite graph and mention what the slope of the line equals.**



5 **What** are the conditions required for an npn transistor to work as a closed switch (on)?

6 A transistor was connected to an ohmmeter where the reading of the device was as shown in the opposite figure, then **what** is the type of the transistor (npn or pnp)? **With explanation**



7 When using an npn transistor in a common emitter amplifier circuit, the collector current was 99 mA and the product of the current transfer ratio (α_e) and the current gain (β_e) was 98.01, **calculate:**

(a) The base current (I_B).

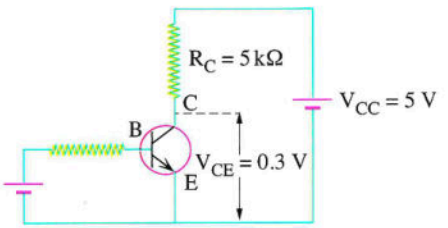
(b) The emitter current (I_E).

8 In a common emitter transistor circuit, **prove that:** $I_B = I_C\left(\frac{1}{\alpha_e} - 1\right)$

9 **In the opposite transistor circuit, if the current gain is 30, calculate:**

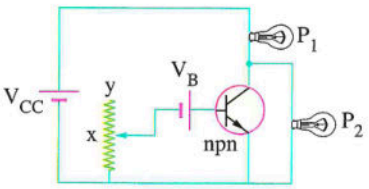
(a) The base current (I_B).

(b) The current transfer ratio (α_e).



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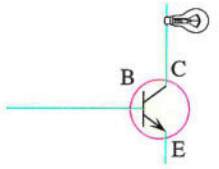
10 The opposite diagram shows an npn transistor circuit working as a closed switch that contains two identical bulbs P_1 and P_2:



(a) **Explain what happens to** the brightness of the two bulbs when moving the slider from point x in the direction of point y.

(b) **Plot the graph** that represents the relation between the voltage (V_1) across the terminals of bulb P_1 and the voltage (V_2) across the terminals of bulb P_2.

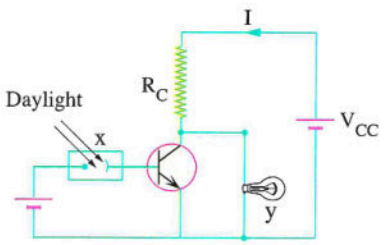
11 The opposite figure represents a transistor connected in series with a small bulb that works on a direct potential difference:



(a) **Complete the diagram** of the electric circuit to light up the bulb.

(b) **What** modification would you make to the circuit in the previous case to prevent the bulb from lighting?

12 In the opposite electronic circuit, a photocell (x) is connected to the base of an npn transistor whose emitter and collector terminals are connected to a lightbulb (y), **explain why** the lightbulb glows when the daylight is gone.



Digital and analog electronic devices

13 **Give reasons for :**

(a) The presence of some defects in sound and image in analog transmission.

(b) Digital electronics are preferred over analog electronics in the electronic devices.

14 **What is the scientific principle on which** digital electronics are based?

15 **Find the equivalent binary number for each of the following decimal numbers:**

(a) 59 (b) 120

16 **Find the corresponding decimal number for each of the following binary numbers:**

(a) $(100110)_2$ (b) $(10011011)_2$

17 **Find** both of the decimal and binary numbers of the quotient of dividing these two binary numbers $\frac{(11110)_2}{(110)_2}$

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18 **Deduce the truth table for:**

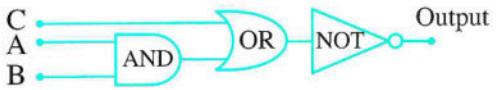
(a) AND gate of two inputs followed by an inverter.

(b) OR gate of two inputs followed by an inverter.

19 **Complete the truth table for each of the following electronic circuits and convert the output result into decimal number:**

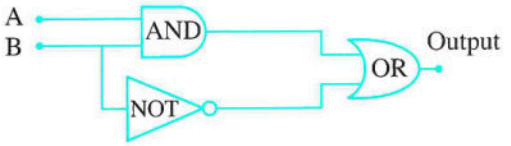
(a)

(A)	(B)	(C)	Output
1	1	0	.....
1	0	1	.....
0	0	1	.....



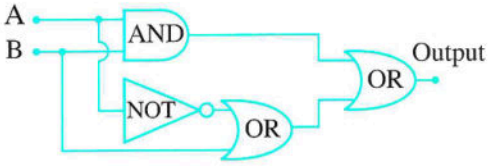
(b)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



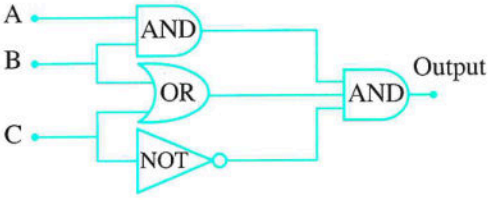
(c)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



(d)

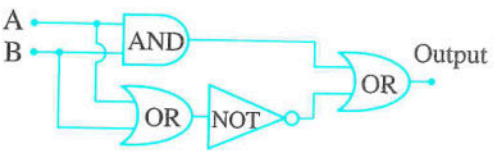
(A)	(B)	(C)	Output
0	0	0	.....
1	1	0	.....
1	0	1	.....
0	1	1	.....
0	0	1	.....
1	1	1	.....



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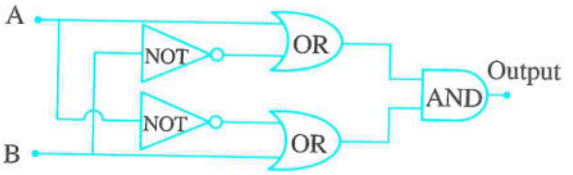
(e)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....



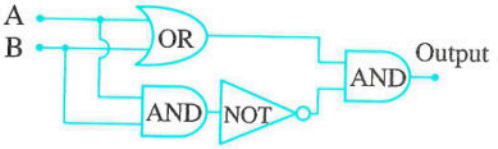
(f)

(A)	(B)	Output
0	0	.....
0	1	.....
1	0	.....
1	1	.....

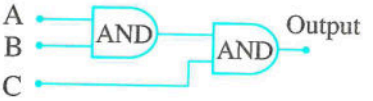


20 **Write the truth table of the following logic circuits :**

(a)



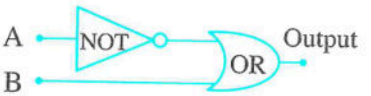
(b)



(c)



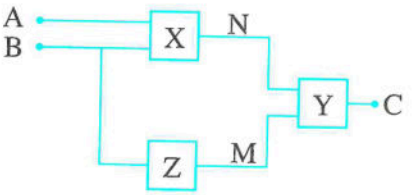
(d)



21 **From the following truth table:**

(a) **Conclude** the types of gates X, Y, Z.

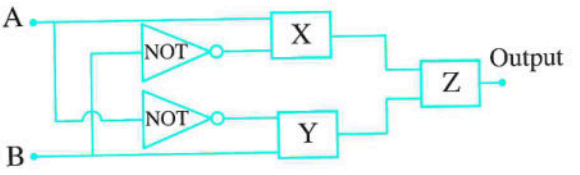
(b) **Complete** the table:



Input				Output
A	B	N	M	C
0	1	1	0	0
1	1	.....	0	.....
1	0	1	.....	1

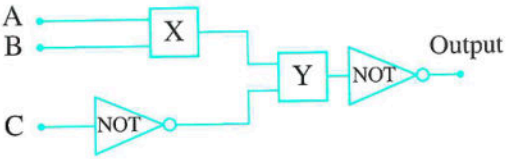
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22 **From the following truth table, conclude the types of gates (X,Y,Z):**



(A)	(B)	Output
0	0	0
0	1	1
1	0	1
1	1	0

23 **The following truth table represents some of input and output values for the logic gates circuit shown in the figure:**



(a) **Identify** the type of gate (X) and gate (Y).

(b) **Find** the output (Z) in the table.

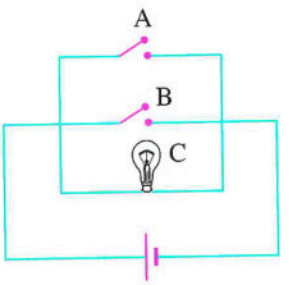
(A)	(B)	(C)	Output
1	1	1	0
0	1	1	1
0	0	0	Z

24 The shown electric circuit is equivalent to a number of logic gates where the switches A and B represent the inputs and the light of the lamp C represents the output.

(a) **Complete** the opposite truth table according to the circuit.

(b) **Draw** the logic gates circuit which fulfill the opposite truth table.

Input		Output (C)
(A)	(B)	
1	1	.....
0	1	.....
1	0	.....
0	0	.....

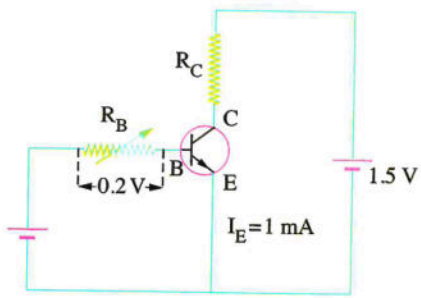


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Questions of the Previous Years Exams on Lesson 2

The transistor

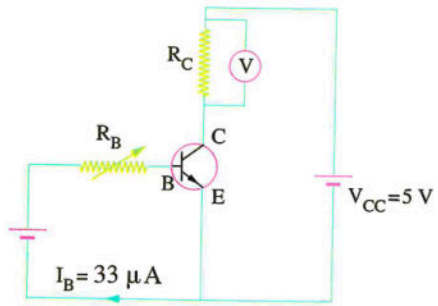
1 The circuit represents a transistor as an inverter gate, if the output potential (V_{CE}) = 0.8 V when the resistance of the base circuit (R_B) = 4000 Ω, the value of the collector circuit resistance (R_C) = ... approximately. *(Exp. June 21)*



(a) 7.37×10^2 Ω (b) 73.7×10^2 Ω

(c) 0.737×10^2 Ω (d) 7370×10^2 Ω

2 The circuit represents a transistor as an amplifier, if the voltmeter reads 4.8 V and the value of R_C = 4.5 kΩ, the values of β_e and α_e respectively are ..... . *(Exp. June 21)*



(a) 32.32, 0.97 (b) 33.67, 0.95

(c) 99, 0.99 (d) 3, 0.75

3 If the base current in npn transistor = 2 mA and α_e = 0.97, the collector current = ...

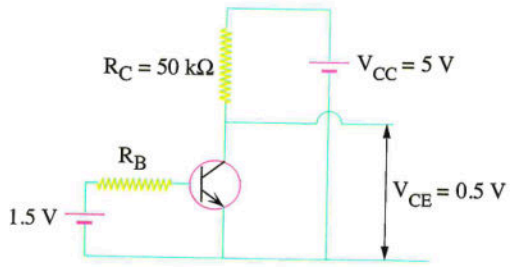
(a) 64.67 mA (b) 1.97 mA (c) 10 mA (d) 50.67 mA

(1st session 21, Practice 25)

4 An npn transistor is used as a current amplifier, if the base current equals 1 mA and the current gain (β_e) is 200, so the collector current equals ..... . *(1st session 21)*

(a) 0.02 A (b) 0.2 A (c) 2 A (d) 20 A

5 In npn transistor circuit, the resistance connected to its collector R_C = 50 kΩ, if its current gain β_e = 30, from the data on the figure, the intensity of base current (I_B) = ... *(2nd session 21)*



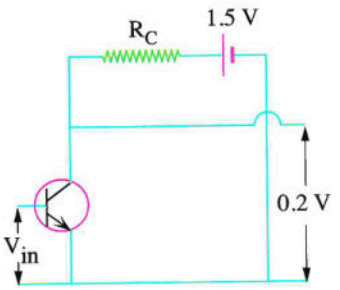
(a) 3×10^{-6} A (b) 9.3×10^{-5} A

(c) 9×10^{-5} A (d) 8.7×10^{-6} A

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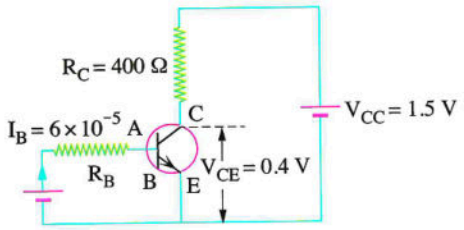
6 A transistor was used as a switch, the output potential (V_{CE}) = 0.2 V and the battery voltage in the collector circuit was 1.5 V, so the potential difference across the resistor in the collector circuit (R_C) = ...

(a) 1.7 V (b) 1.3 V
(c) 0.3 V (d) 7.5 V *(2nd session 21)*



7 The figure shows a circuit of transistor (npn) as amplifier, so the ratio of $\frac{\alpha_e}{\beta_e}$ = ...

(a) 2.7×10^{-3} (b) 2.1×10^{-2}
(c) 1.1×10^{-2} (d) 2.8×10^{-3}
(1st session 22)



8 If the current division (transfer) ratio of a transistor $\alpha_e = 0.99$ then, the ratio of :

$$\frac{\text{the current of the emitter } I_E}{\text{the current of the base } I_B} = \text{.....}$$

(2nd session 22)

(a) 100 (b) 99 (c) 200 (d) 198

9 If the base current in an npn transistor is 6 μA and ($\alpha_e = 0.95$), then the currents through the emitter and the collector, respectively, are ..... .
(Exp. 23)

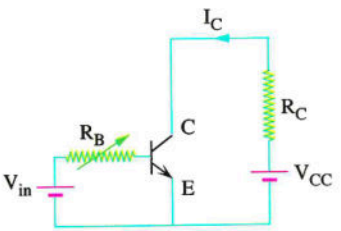
	I_E	I_C
(a)	120 μA	114 μA
(b)	114 μA	120 μA
(c)	12 μA	11.4 μA
(d)	242 μA	240 μA

10 In the transistor circuit, if the current of the emitter is 120 times as that of the base, then α_e equals .....
(1st session 23)

(a) 0.96 (b) 120 (c) 119 (d) 0.99

11 The electric circuit shown in the figure represents a transistor used as a switch, so on increasing the value of R_B to double its value. Which of the following choices correctly describes what happens to the value of V_{CC}? *(2nd session 23)*

(a) It remains constant.
(b) It decreases to half its value.
(c) It increases to double its value.
(d) It becomes equals to zero.



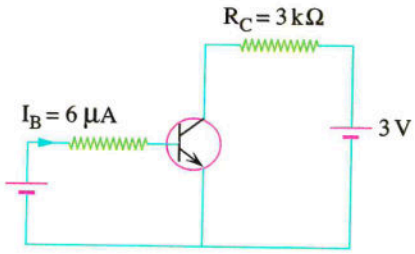
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12 The current gain (β_e) for a transistor = 150, so (α_e) = ... *(2nd session 23)*

(a) 1.01 (b) 0.99

(c) 0.97 (d) 1.1

13 The opposite figure shows npn transistor ($\beta_e = 99$) so, ... *(1st session 24)*



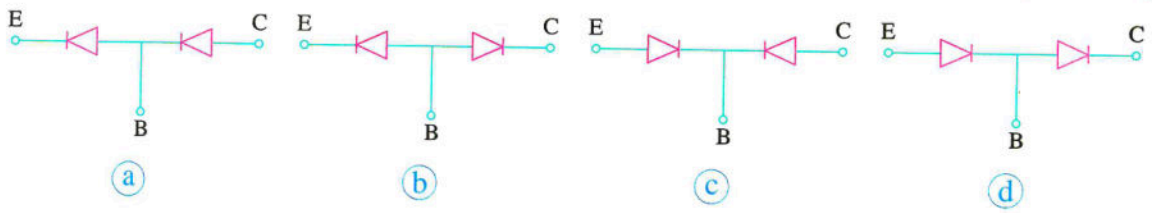
	Output voltage (V_{CE})	Collector current I_C
(a)	2.982 V	0.06 μA
(b)	1.782 V	16.5 μA
(c)	1.218 V	594 μA
(d)	2.982 V	16.5 μA

14 If the current gain (β_e) in a transistor = 93.6, then the ratio $\left(\frac{I_E}{I_B}\right)$ = ... *(2nd session 24)*

(a) 93.6 (b) 95.6

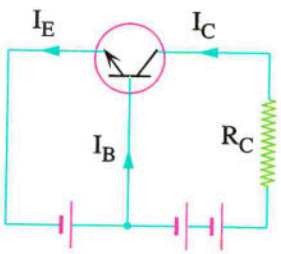
(c) 94.6 (d) 92.6

15 The npn transistor can be considered as two diodes connected together. Which of the following figures is the correct figure representing the npn transistor? *(Practice 25)*



(a) (b) (c) (d)

16 A transistor is connected as in the opposite figure. Which of the following choices describes correctly the use of transistor in that circuit? *(Practice 25)*



(a) Amplifies both of the electric power and the electric current intensity.

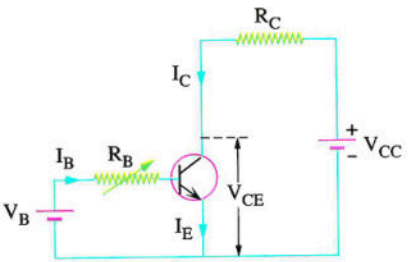
(b) Amplifies the electric power but does not amplify the electric current intensity.

(c) Amplifies the electric current intensity but does not amplify the electric power.

(d) Neither amplifies the electric power nor the electric current intensity.

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17 The figure shows npn transistor circuit.
What is the effect of increasing the value of the variable resistor (R_B) on each of the collector current (I_C) and the output voltage (V_{CE})?

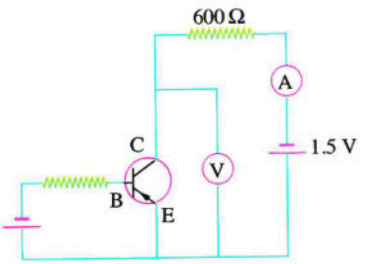


	Collector current (I_C)	Output voltage (V_{CE})
a	Increases	Increases
b	Decreases	Increases
c	Does not change	Decreases
d	Decreases	Does not change

(1st session 25)

18 In the transistor circuit shown in the figure, the ammeter reads 2 mA. So, the voltmeter reads ..... .

a) 0.9 V b) 0.3 V
c) 0.6 V d) 0.5 V *(2nd session 25)*



19 In a transistor circuit in which $V_{CC} = 1.5$ V, the potential difference between the collector and the emitter (V_{CE}) = 0.5 V and the resistance of the collector circuit (R_C) = 500 Ω, **calculate** the collector current (I_C). *(Practice 25)*

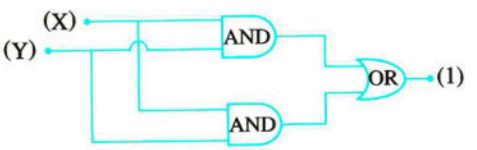
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.....

Digital and analog electronic devices

20 A group of logic gates has an output voltage (1), as shown in the figure which of the probabilities shown in the table verifies that? *(Exp. June 21)*

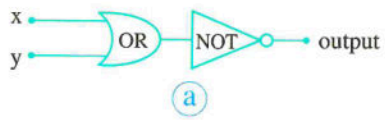


	X	Y
a	0	0
b	1	0
c	1	1
d	0	1

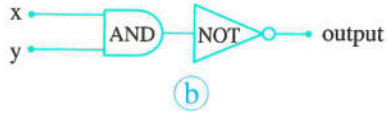
CamScanner

21 Which group of the following logic gates verifies the shown truth table?

input		output
x	y	
1	0	1

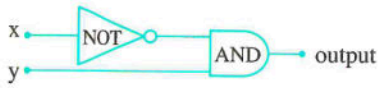


(a)

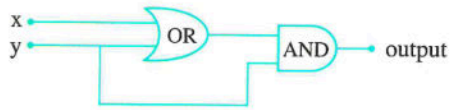


(b)

(1st session 21)



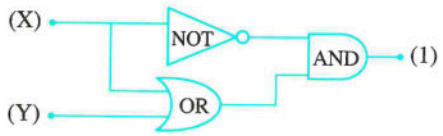
(c)



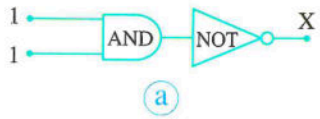
(d)

22 The group of logic gates in the figure has an output voltage (1), which of the following choices in the table for the inputs potentials verifies that? *(2nd session 21, 1st session 22)*

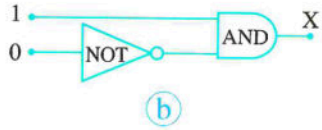
	X	Y
(a)	0	0
(b)	1	0
(c)	1	1
(d)	0	1



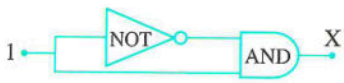
23 Which logic circuit has a high output voltage at point X? *(2nd session 22)*



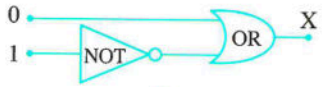
(a)



(b)



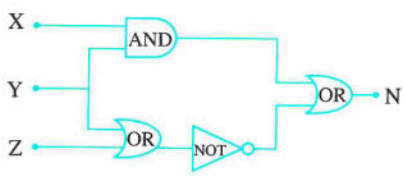
(c)



(d)

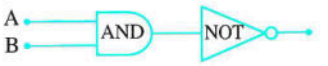
24 In the illustrated circuit of logic gates, which of the following choices gives an output of N = 0? *(Exp. 23)*

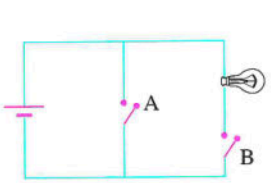
	X	Y	Z
(a)	0	1	0
(b)	1	1	0
(c)	0	0	0
(d)	1	0	0



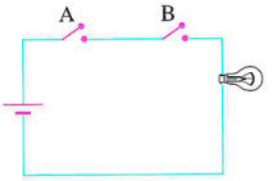
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25 Which of the following simple electric circuits represents the combination of the opposite logic gates? *(1st session 23)*

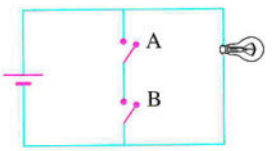




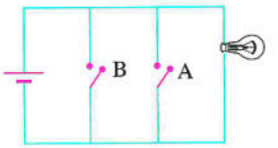
(a)



(b)



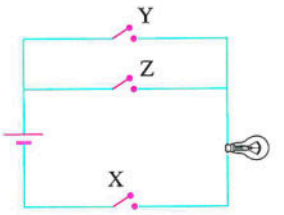
(c)

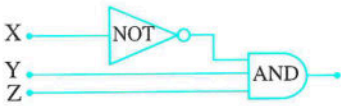


(d)

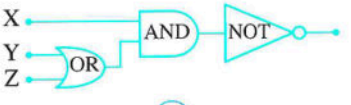
26 The figure shows a simple electric circuit that contains three switches (X , Y and Z) working as inputs to a logic circuit, while the lamp gives the output. The group of logic gates that is represented by this simple electric circuit is ..... .

(2nd session 23)

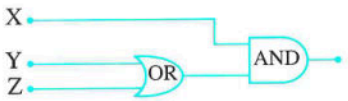




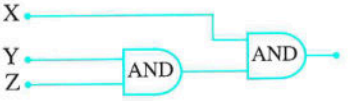
(a)



(b)



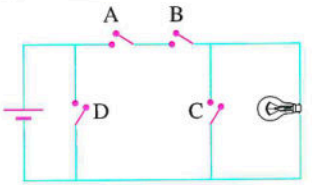
(c)

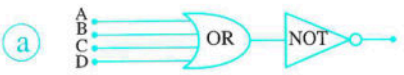


(d)

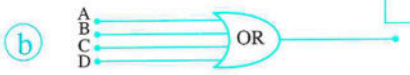
27 The opposite electric circuit is equivalent to the logic gate of shape ..... .

(1st session 24)

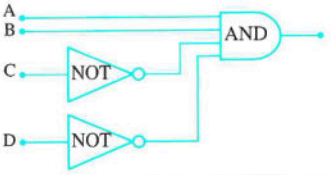




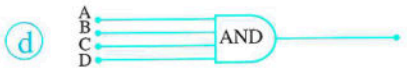
(a)



(b)

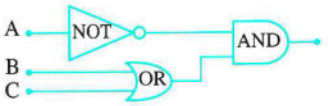


(c)



(d)

28 The opposite figure represents several logic gates connected together: Which choice produces high output voltage?



	A	B	C
(a)	0	0	0
(b)	0	0	1
(c)	1	1	0
(d)	1	1	1

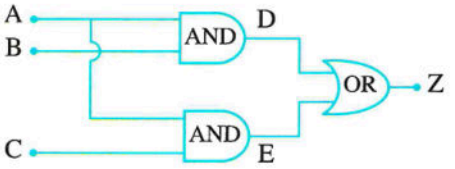
(2nd session 24)

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29 The logic gate whose equivalent electric circuit has two switches connected in parallel is ..... *(Practice 25)*

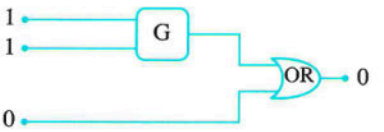
(a) NOT (b) AND (c) OR (d) NOT or AND

30 A group of logic gates are connected together as in the opposite figure, which of the following choices can be obtained from that group? *(Practice 25)*



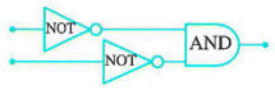
	A	B	C	Z
(a)	0	1	1	1
(b)	1	0	1	0
(c)	1	1	0	0
(d)	0	1	1	0

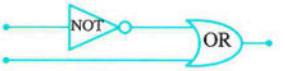
31 The opposite figure shows a circuit containing a group of logic gates where the output is (Low) when the input is as shown, so (G) may be ..... *(Practice 25)*



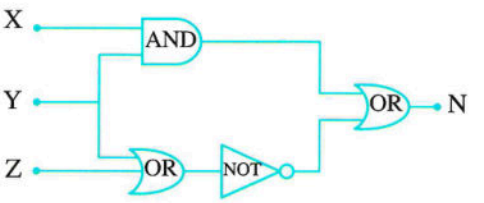
(a) 

(b) 

(c) 

(d) 

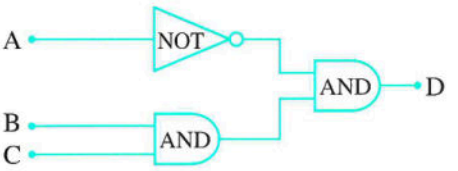
32 The figure shows a set of logic gates connected together. Which of the following choices for the input values makes the output (N) equal to zero? *(1st session 25)*



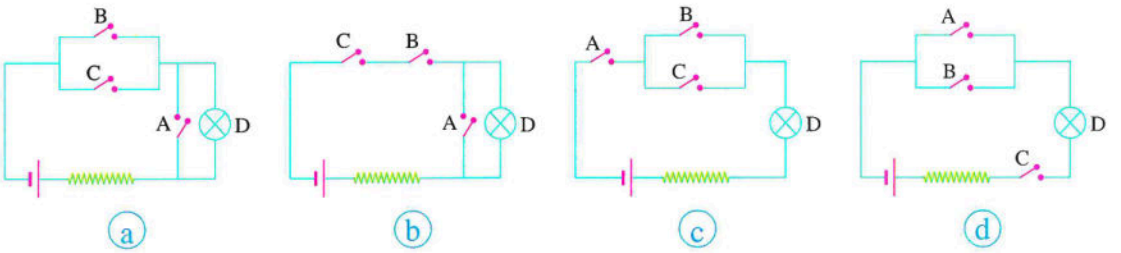
	X	Y	Z
(a)	0	1	0
(b)	1	1	0
(c)	1	1	1
(d)	0	0	0

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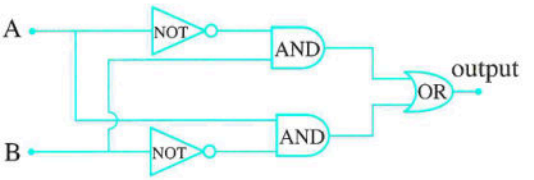
33 The figure shows a set of logic gates connected together. Which of the following electrical circuits represents the operation of this set of gates?



(2nd session 25)



34 The diagram shows some logic gates connected to each other. The number of possibilities for the output to be high is .....



(a) 0

(b) 1

(c) 2

(d) 3

(Practice 26)

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Test on Chapters 7 and 8

Answered

First **Choose the correct answer:**

1 When the temperature of pure semiconductor increases, its electric conductivity ..... .

(a) decreases because of the decrease of charge carriers
(b) decreases because of the increase of charge carriers
(c) increases because of the increase of charge carriers
(d) increases because of the decrease of charge carriers

2 What property do laser rays have that makes them usable in guiding missiles and airplanes?

(a) High speed. (b) Spectral purity.
(c) Having short wavelengths. (d) Being parallel and concentrated.

3 All laser devices consist of three basic elements which are:

(I) effective medium **(II)** power source **(III)** resonant cavity

And the argon laser differs from the ruby laser in the nature of ..... .

(a) element (I) only (b) both elements (I), (II) only
(c) both elements (I), (III) only (d) all elements (I), (II), (III)

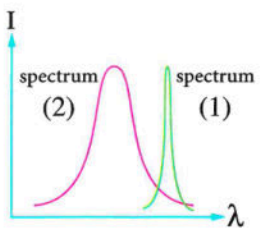
4 In either analog or digital transmitting devices, ...

(a) electric signals are converted into analog electromagnetic waves
(b) electric signals are converted into digital electromagnetic waves
(c) physical messages are converted into digital electric signals
(d) physical messages are converted into analog electric signals

5 Which of the following rays **doesn't** represent electromagnetic radiation?

(a) Laser rays. (b) X-rays. (c) Cathode rays. (d) Blackbody radiation.

6 Two spectra (1) and (2), each of them is monochromatic and they are analyzed by a spectrometer. The opposite figure represents the relation between radiation intensity (I) and wavelength (λ) for both spectra, so which of the following statements can be considered a suitable conclusion?

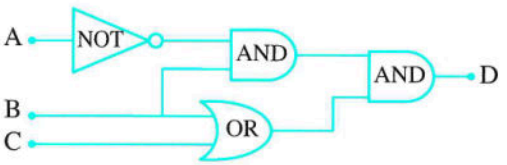


(a) The capability of spectrum (1) to scatter is higher than spectrum (2).
(b) A photon of spectrum (1) has higher energy than that of spectrum (2).
(c) Spectrum (1) propagates in space faster than spectrum (2).
(d) Spectrum (1) has coherent photons.

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7 In the opposite logic circuit, which of the following inputs gives the output D = 1?

	A	B	C
(a)	0	0	1
(b)	1	0	0
(c)	0	1	0
(d)	1	0	1



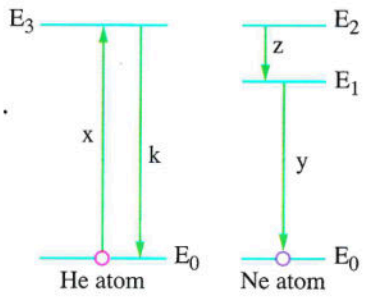
8 The opposite figure illustrates some transitions x, y, z and k in the helium-neon laser. The correct arrangement for these transitions according to the order of their occurrence is ..... .

(a) k ⟶ y ⟶ z ⟶ x

(b) x ⟶ y ⟶ z ⟶ k

(c) x ⟶ k ⟶ z ⟶ y

(d) z ⟶ y ⟶ x ⟶ k



9 Number 40 in the decimal system is equivalent in the binary system to ..... .

(a) $(110000)_2$ (b) $(101000)_2$ (c) $(001010)_2$ (d) $(001100)_2$

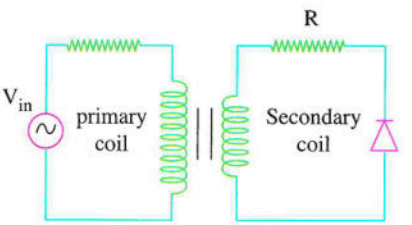
10 In the opposite circuit, what is the type of current through resistor R?

(a) Direct current.

(b) Half-wave rectified current.

(c) Alternating current.

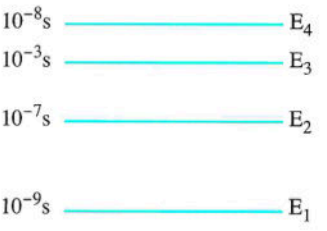
(d) Unidirectional of variable intensity current.



11 The opposite figure illustrates multiple excitation levels for an atom of an element and the lifetime of each level, hence the atoms of this element can reach population inversion when they are excited to level ..... .

(a) E_1 (b) E_2

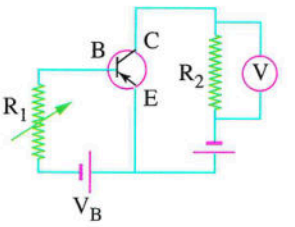
(c) E_3 (d) E_4



12 In the opposite transistor circuit, what happens to the voltmeter reading when increasing the resistance of R_1?

(a) Increases. (b) Decreases.

(c) Becomes zero. (d) Doesn't change.

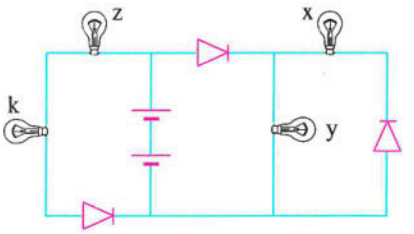


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13 The radiation produced by stimulated emission is distinguished from its counterpart produced by spontaneous emission in all the following characteristics **except** that ..... .

(a) its photons travel in one direction
(b) its waves are in phase
(c) its photons have one wavelength
(d) its photons have a greater speed

14 The opposite electric circuit contains a battery, multiple identical lightbulbs and multiple identical diodes, so the lightbulb that glows with the greatest brightness is ..... .

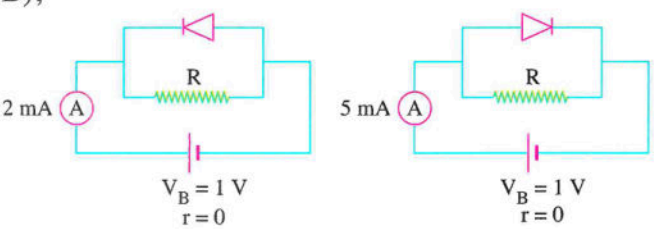


(a) x
(b) y
(c) z
(d) k

15 The drift current in a pn junction is due to the motion of ..... .

(a) positive ions from the n-region to the p-region
(b) free electrons from the p-region to the n-region
(c) holes from the p-region to the n-region
(d) negative ions from the p-region to the n-region

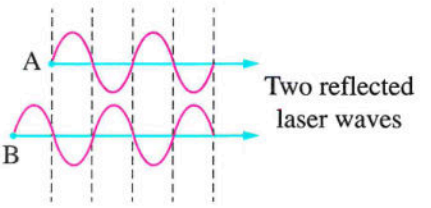
16 In the opposite two electric circuits (A, B), assuming that the resistance of the diode in the case of reverse biasing is infinite, so the resistance of R and the resistance of the diode in the case of forward biasing, respectively, are ..... .



Circuits (A) **Circuits (B)**

(a) 300 Ω, 333.33 Ω
(b) 300 Ω, 155.33 Ω
(c) 500 Ω, 333.33 Ω
(d) 500 Ω, 155.33 Ω

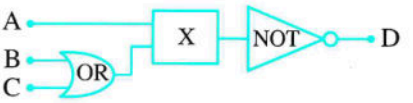
17 The opposite figure illustrates two laser waves reflected from two points A and B lying on a body during 3D hologram recording, if the path difference between them is 3164 Å, so the wavelength of the laser equals ..... .



(a) 1582 Å
(b) 3164 Å
(c) 6328 Å
(d) 9492 Å

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18 The opposite figure illustrates logic gates circuit where an unknown logic gate (x) is one of them, so according to the logic circuit and its truth table, logic gate X is ..... .



A	B	C	D
0	0	0	1
1	0	1	0
0	1	1	1
1	1	1	0

ⓐ AND

ⓑ OR

ⓒ NOT

ⓓ OR or AND

19 If the concentration of holes or that of free electrons in a pure semiconductor crystal is 2×10^{8} cm^{-3} and when atoms of a certain element are added into it, the concentration of the holes inside it has increased to 4×10^{10} cm^{-3}, then the crystal type and the concentration of free electrons are ..... .

ⓐ n-type, 10^{6} cm^{-3}

ⓑ p-type, 2×10^{8} cm^{-3}

ⓒ n-type, 2×10^{8} cm^{-3}

ⓓ p-type, 10^{6} cm^{-3}

Second Answer the following question

20 The opposite figure illustrates a transistor circuit, then:

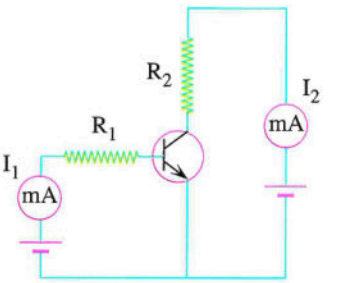
(1) Mention one use of this circuit.

(2) If $\alpha_e = 0.98$, **find the ratio of** $\left(\frac{I_2}{I_1}\right)$.

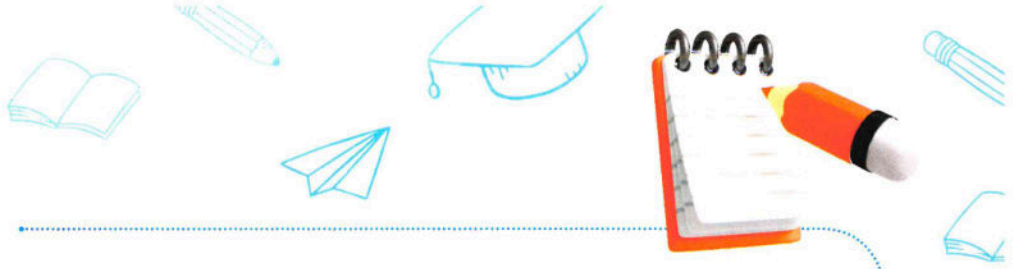
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